

EAST TEXAS, TX, USA

WMO: 722470

Lat: 32.385N

Lon: 94.712W

Elev: 111

StdP: 100.00

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.1	-11.1	1.5	1.5	-8.7	1.8	3.1	10.8	13.6	9.5	15.8	2.1	350	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	37.7	24.0	36.4	24.2	35.0	24.3	26.3	32.4	25.8	31.8	25.5	31.4	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.4	28.1	24.5	19.7	27.8	24.0	19.2	27.6	82.2	32.5	80.4	31.9	78.7	31.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.1	7.3	DB	-7.1	39.1	2.1	1.7	-8.6	40.3	-9.8	41.3	-11.0	42.3	-12.5	43.6	
			WB	-8.2	26.9	2.0	0.7	-9.6	27.4	-10.8	27.8	-11.9	28.1	-13.3	28.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	8.6	10.8	14.9	18.9	23.2	27.1	28.8	28.9	25.6	19.6	13.9	9.5
	DBStd	8.39	5.57	5.71	5.00	4.11	3.36	2.29	2.05	2.33	3.19	4.46	5.13	5.28
	HDD10.0	262	94	55	15	1	0	0	0	0	0	1	21	76
	HDD18.3	1166	304	218	128	42	6	0	0	0	1	39	149	278
	CDD10.0	3618	52	77	167	269	409	513	581	586	468	298	138	60
	CDD18.3	1481	3	7	22	59	157	263	323	328	219	79	17	4
	CDH23.3	15669	9	33	140	472	1377	2802	3837	3979	2218	693	93	16
CDH26.7	7374	0	2	18	96	494	1332	1988	2158	1048	224	12	1	
Wind	WSAvg	3.2	3.4	3.7	3.7	3.9	3.5	3.1	2.8	2.6	2.4	2.8	3.1	3.3
Precipitation	PrecAvg	1198	109	100	112	89	102	111	84	75	94	105	98	115
	PrecMax	1788	218	237	239	329	229	370	280	187	232	380	283	255
	PrecMin	723	18	10	13	13	21	3	5	1	26	3	9	17
	PrecStd	285	52	54	51	66	58	85	58	45	58	86	63	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.4	28.7	31.3	34.2	37.7	38.9	40.2	37.9	32.7	28.3	25.3
		MCWB	17.0	17.8	19.3	21.8	23.1	24.1	24.2	23.5	22.8	22.4	20.8	19.1
	2%	DB	22.6	24.2	26.9	29.0	32.3	35.9	37.5	37.9	35.4	31.0	26.0	22.9
		MCWB	16.5	17.5	19.0	20.8	23.2	24.1	24.1	23.9	23.5	21.9	19.6	18.4
	5%	DB	20.6	22.2	24.8	27.6	31.0	34.1	35.9	36.7	33.8	29.2	23.9	21.1
		MCWB	16.2	17.1	17.7	20.1	22.9	24.1	24.5	24.4	23.3	21.1	18.4	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	18.5	20.3	22.9	26.3	29.6	32.8	34.3	35.2	32.3	27.4	22.3	18.9
		MCWB	14.6	16.0	17.0	19.2	22.4	24.3	24.7	24.4	23.1	20.2	18.1	15.6
	0.4%	WB	19.7	20.8	21.2	23.8	25.5	26.7	26.9	26.8	26.3	25.1	22.7	20.9
		MCDB	22.8	23.3	25.4	27.5	30.6	32.0	32.6	32.8	31.6	29.4	25.8	23.0
	2%	WB	18.5	19.5	20.3	22.5	24.6	26.0	26.3	26.3	25.5	24.1	21.1	19.6
		MCDB	20.9	22.4	24.2	26.8	29.6	31.4	32.4	32.6	30.7	28.1	24.0	22.2
Mean Daily Temperature Range	5% DB	WB	17.4	18.3	19.5	21.6	23.9	25.5	25.8	25.8	24.9	23.1	20.1	18.2
		MCDB	19.7	21.1	23.1	25.9	28.8	31.0	31.8	32.1	30.0	26.7	22.9	20.5
	5% WB	WB	15.2	16.8	18.3	20.8	23.3	25.0	25.5	25.4	24.4	22.1	18.8	16.3
		MCDB	17.6	19.6	21.9	24.7	27.9	30.5	31.4	31.6	29.3	25.4	21.7	18.5
Clear-Sky Solar Irradiance	taub	0.325	0.340	0.360	0.398	0.429	0.452	0.467	0.462	0.418	0.363	0.349	0.330	
	taud	2.504	2.468	2.424	2.321	2.287	2.264	2.258	2.263	2.348	2.479	2.466	2.501	
	Ebn at Noon	897	916	919	893	863	839	824	824	849	879	859	870	
	Edh at Noon	87	99	111	128	134	137	137	134	119	97	89	83	
All-Sky Solar Radiation	RadAvg	2.74	3.07	4.30	5.36	5.83	6.29	6.36	5.96	5.07	4.08	2.96	2.38	
	RadStd	0.25	0.51	0.40	0.33	0.44	0.52	0.47	0.48	0.45	0.55	0.35	0.31	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+179	+132
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page